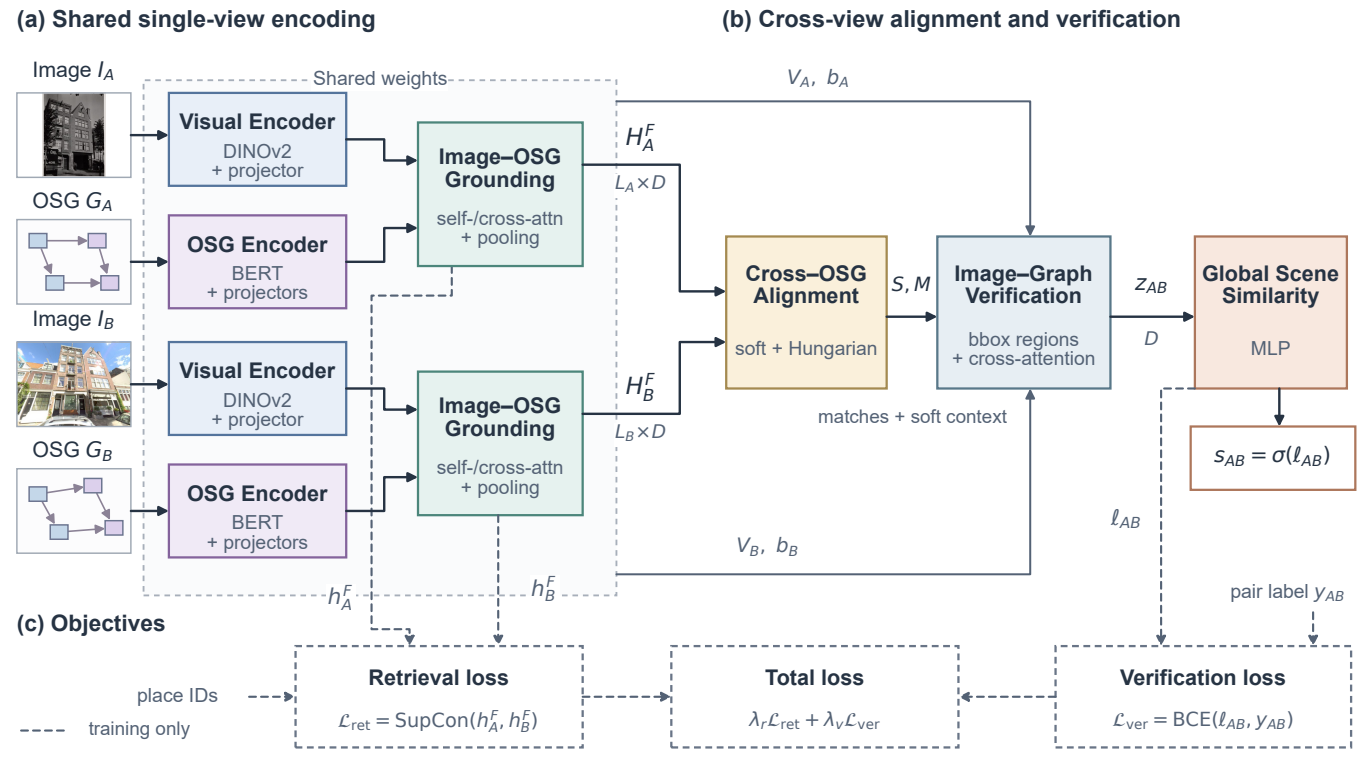

(a) Shared single-view encoding
(b) Cross-view alignment and verification
Image I_A
OSG G_A
Image I_B
OSG G_B
Shared weights
Visual Encoder
DINOv2
+ projector
OSG Encoder
BERT
+ projectors
Visual Encoder
DINOv2
+ projector
OSG Encoder
BERT
+ projectors
Image–OSG Grounding
self-/cross-attn
+ pooling
Image–OSG Grounding
self-/cross-attn
+ pooling
H_A^F
$L_A \times D$
H_B^F
$L_B \times D$
$V_A,\ b_A$
$V_B,\ b_B$
Cross–OSG Alignment
soft + Hungarian
S, M
Image–Graph Verification
bbox regions
+ cross-attention
z_{AB}
D
Global Scene Similarity
MLP
$s_{AB} = \sigma(\ell_{AB})$
matches + soft context
ℓ_{AB}
h_A^F
h_B^F
(c) Objectives
pair label y_{AB}
place IDs
training only
Retrieval loss
$\mathcal{L}_{\text{ret}} = \text{SupCon}(h_A^F, h_B^F)$
Total loss
$\lambda_r \mathcal{L}_{\text{ret}} + \lambda_v \mathcal{L}_{\text{ver}}$
Verification loss
$\mathcal{L}_{\text{ver}} = \text{BCE}(\ell_{AB}, y_{AB})$